WEEK 9: SALES AND CUSTOMER ANALYTICS SYSTEM

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```
SQL> SELECT COUNT(*) AS TOTAL_ORDERS
  2  FROM ORDERS;

TOTAL_ORDERS
------------
          10
```

```
SQL> SELECT SUM(TOTAL_AMOUNT) AS TOTAL_REVENUE
  2  FROM ORDERS;

TOTAL_REVENUE
-------------
         7589
```

```
         7589

SQL> SELECT AVG(TOTAL_AMOUNT) AS AVERAGE_ORDER_VALUE
  2  FROM ORDERS;

AVERAGE_ORDER_VALUE
-------------------
              758.9
```

```
SQL> SELECT MAX(TOTAL_AMOUNT) AS HIGHEST_ORDER_AMOUNT,
  2         MIN(TOTAL_AMOUNT) AS LOWEST_ORDER_AMOUNT
  3  FROM ORDERS;

HIGHEST_ORDER_AMOUNT LOWEST_ORDER_AMOUNT
-------------------- -------------------
                1299                 299
```

```
SQL> SELECT SUM(TOTAL_AMOUNT) AS TOTAL_SALES
  2  FROM ORDERS
  3  WHERE ORDER_DATE >= DATE '2026-09-01'
  4  AND ORDER_DATE < DATE '2026-10-01';

TOTAL_SALES
-----------
       7589
```

2.GENERATE TOTAL SALES REPORT

1. Total number of orders placed

```
SQL> SELECT COUNT(ORDER_ID) AS TOTAL_ORDERS
  2  FROM ORDERS;

TOTAL_ORDERS
------------
          10
```

2.Total revenue generated from sales

```
SQL> SELECT SUM(TOTAL_AMOUNT) AS TOTAL_REVENUE
  2  FROM ORDERS;

TOTAL_REVENUE
-------------
         7589
```

3.Average order value

```
SQL> SELECT AVG(TOTAL_AMOUNT) AS AVERAGE_ORDER_VALUE
  2  FROM ORDERS;

AVERAGE_ORDER_VALUE
-------------------
              758.9
```

4.highest and lowest order amount

```
SQL> SELECT MAX(TOTAL_AMOUNT) AS HIGHEST_ORDER_AMOUNT,
  2         MIN(TOTAL_AMOUNT) AS LOWEST_ORDER_AMOUNT
  3  FROM ORDERS;

HIGHEST_ORDER_AMOUNT LOWEST_ORDER_AMOUNT
-------------------- -------------------
                1299                 299
```

5.total sales for a specfic period

Example; september 20 to september 30 , 2026

```
SQL> SELECT SUM(TOTAL_AMOUNT) AS TOTAL_SALES
  2  FROM ORDERS
  3  WHERE ORDER_DATE >= DATE '2026-09-20'
  4  AND ORDER_DATE < DATE '2026-10-01';

TOTAL_SALES
-----------
       4095
```

3.customer purchase analysis

1.total number of orders placed by each customer

2 total amount spent by each customer

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.CUSTOMER_NAME,
  3         COUNT(O.ORDER_ID) AS TOTAL_ORDERS
  4  FROM CUSTOMER C
  5  LEFT JOIN ORDERS O
  6  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  7  GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
  8  ORDER BY TOTAL_ORDERS DESC;

CUSTOMER_I CUSTOMER_NAME                                      TOTAL_ORDERS
---------- -------------------------------------------------- ------------
C002       Ananya                                                        2
C001       Rahul                                                         2
C004       Meena                                                         2
C005       Karthik                                                       2
C003       Vikram                                                        2
```

3.average spending per customer

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.CUSTOMER_NAME,
  3         NVL(SUM(O.TOTAL_AMOUNT), 0) AS AMOUNT_SPENT
  4  FROM CUSTOMER C
  5  LEFT JOIN ORDERS O
  6  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  7  GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
  8  ORDER BY AMOUNT_SPENT DESC;

CUSTOMER_I CUSTOMER_NAME                                       AMOUNT_SPENT
---------- -------------------------------------------------- ------------
C004       Meena                                                       1947
C005       Karthik                                                     1798
C003       Vikram                                                      1598
C002       Ananya                                                      1298
C001       Rahul                                                        948

SQL> SELECT C.CUSTOMER_ID
```

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.CUSTOMER_NAME,
  3         NVL(AVG(O.TOTAL_AMOUNT), 0) AS AVERAGE_SPENDING
  4  FROM CUSTOMER C
  5  LEFT JOIN ORDERS O
  6  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  7  GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
  8  ORDER BY AVERAGE_SPENDING DESC;

CUSTOMER_I CUSTOMER_NAME
---------- ----------------------------------------------
C004       Meena
C005       Karthik
C003       Vikram
C002       Ananya
C001       Rahul
```

4.customer with maximum purchase amount

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.CUSTOMER_NAME,
  3         SUM(O.TOTAL_AMOUNT) AS TOTAL_SPENDING
  4  FROM CUSTOMER C
  5  JOIN ORDERS O
  6  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  7  GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
  8  HAVING SUM(O.TOTAL_AMOUNT) = (
  9      SELECT MAX(TOTAL_SPENT)
 10      FROM (
 11          SELECT SUM(TOTAL_AMOUNT) AS TOTAL_SPENT
 12          FROM ORDERS
 13          GROUP BY CUSTOMER_ID
 14      )
 15  );

CUSTOMER_I CUSTOMER_NAME                                TOTAL_SPENDING
---------- ---------------------------------------------- --------------
C004       Meena                                                 1947
```

5. customers with fewer purchasess

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.CUSTOMER_NAME,
  3         COUNT(O.ORDER_ID) AS TOTAL_ORDERS
  4  FROM CUSTOMER C
  5  LEFT JOIN ORDERS O
  6  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  7  GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
  8  HAVING COUNT(O.ORDER_ID) < 2
  9  ORDER BY TOTAL_ORDERS ASC;

no rows selected
```

4.FIND TOP CUSTOMER BASED ON PURCHASE AMOUNT

1.Top 5 customers based on total spending

```
SQL> SELECT CUSTOMER_ID,
  2         CUSTOMER_NAME,
  3         TOTAL_SPENDING
  4  FROM (
  5      SELECT C.CUSTOMER_ID,
  6             C.CUSTOMER_NAME,
  7             SUM(O.TOTAL_AMOUNT) AS TOTAL_SPENDING
  8      FROM CUSTOMER C
  9      JOIN ORDERS O
 10      ON C.CUSTOMER_ID = O.CUSTOMER_ID
 11      GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
 12      ORDER BY TOTAL_SPENDING DESC
 13  )
 14  WHERE ROWNUM <= 5;

CUSTOMER_I CUSTOMER_NAME                                      TOTAL_SPENDING
---------- -------------------------------------------------- --------------
C004       Meena                                                        1947
C005       Karthik                                                      1798
C003       Vikram                                                       1598
C002       Ananya                                                       1298
C001       Rahul                                                         948
```

2.customers with maximum number of orders

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.CUSTOMER_NAME,
  3         COUNT(O.ORDER_ID) AS TOTAL_ORDERS
  4  FROM CUSTOMER C
  5  JOIN ORDERS O
  6  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  7  GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
  8  HAVING COUNT(O.ORDER_ID) = (
  9      SELECT MAX(ORDER_COUNT)
 10      FROM (
 11          SELECT COUNT(*) AS ORDER_COUNT
 12          FROM ORDERS
 13          GROUP BY CUSTOMER_ID
 14      )
 15  );

CUSTOMER_I CUSTOMER_NAME                                    TOTAL_ORDERS
---------- ---------------------------------------------- ------------
C002       Ananya                                                    2
C001       Rahul                                                     2
C003       Vikram                                                    2
C005       Karthik                                                   2
C004       Meena                                                     2
```

3.Frequnet customers

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.CUSTOMER_NAME,
  3         COUNT(O.ORDER_ID) AS TOTAL_ORDERS
  4  FROM CUSTOMER C
  5  JOIN ORDERS O
  6  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  7  GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
  8  HAVING COUNT(O.ORDER_ID) >= 3
  9  ORDER BY TOTAL_ORDERS DESC;

no rows selected
```

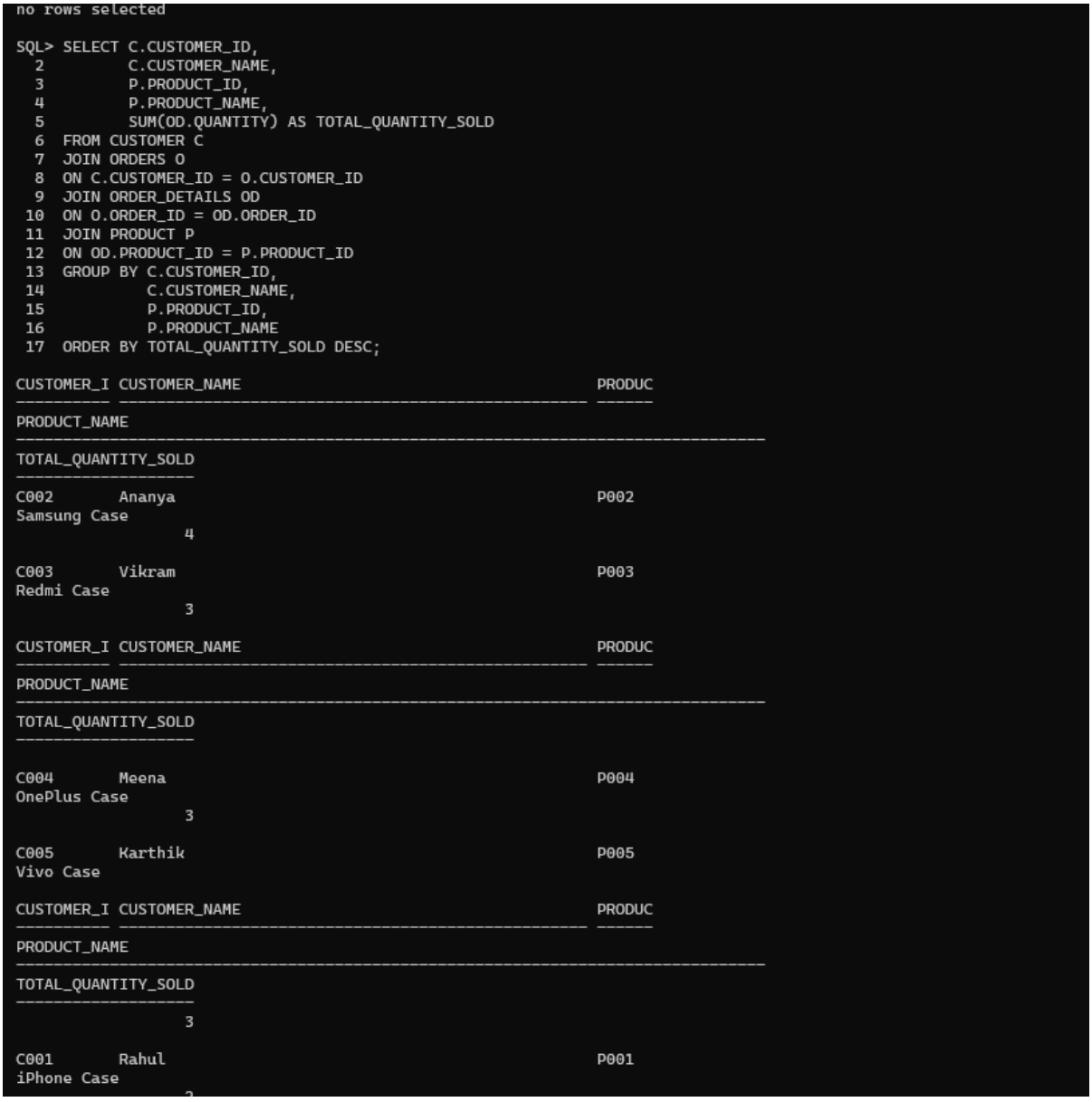
4.high values customers

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.CUSTOMER_NAME,
  3         SUM(O.TOTAL_AMOUNT) AS TOTAL_SPENDING
  4  FROM CUSTOMER C
  5  JOIN ORDERS O
  6  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  7  GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
  8  HAVING SUM(O.TOTAL_AMOUNT) > (
  9      SELECT AVG(CUSTOMER_SPENDING)
 10      FROM (
 11          SELECT SUM(TOTAL_AMOUNT) AS CUSTOMER_SPENDING
 12          FROM ORDERS
 13          GROUP BY CUSTOMER_ID
 14      )
 15  )
 16  ORDER BY TOTAL_SPENDING DESC;

CUSTOMER_I CUSTOMER_NAME                                      TOTAL_SPENDING
---------- -------------------------------------------------- --------------
C004       Meena                                                        1947
C005       Karthik                                                      1798
C003       Vikram                                                       1598
```

5. identify best selling product

1. products with maximum sales quantity



```
no rows selected

SQL> SELECT C.CUSTOMER_ID,
  2         C.CUSTOMER_NAME,
  3         P.PRODUCT_ID,
  4         P.PRODUCT_NAME,
  5         SUM(OD.QUANTITY) AS TOTAL_QUANTITY_SOLD
  6  FROM CUSTOMER C
  7  JOIN ORDERS O
  8  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  9  JOIN ORDER_DETAILS OD
 10  ON O.ORDER_ID = OD.ORDER_ID
 11  JOIN PRODUCT P
 12  ON OD.PRODUCT_ID = P.PRODUCT_ID
 13  GROUP BY C.CUSTOMER_ID,
 14           C.CUSTOMER_NAME,
 15           P.PRODUCT_ID,
 16           P.PRODUCT_NAME
 17  ORDER BY TOTAL_QUANTITY_SOLD DESC;

CUSTOMER_I CUSTOMER_NAME                                      PRODUC
---------- -------------------------------------------------- ------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------
C002       Ananya                                             P002
Samsung Case
                   4

C003       Vikram                                             P003
Redmi Case
                   3

CUSTOMER_I CUSTOMER_NAME                                      PRODUC
---------- -------------------------------------------------- ------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------

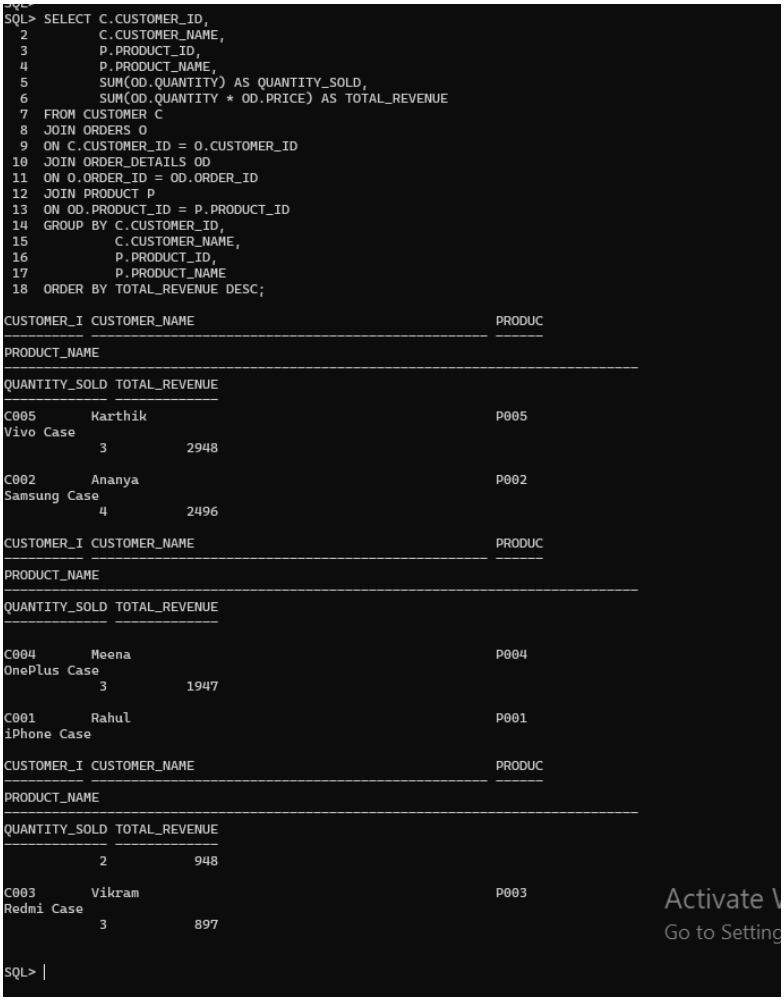
C004       Meena                                              P004
OnePlus Case
                   3

C005       Karthik                                            P005
Vivo Case

CUSTOMER_I CUSTOMER_NAME                                      PRODUC
---------- -------------------------------------------------- ------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------
                   3

C001       Rahul                                              P001
iPhone Case
                   2
```

2.products generating highest revenue



```
SQL> SELECT C.CUSTOMER_ID,
  2         C.CUSTOMER_NAME,
  3         P.PRODUCT_ID,
  4         P.PRODUCT_NAME,
  5         SUM(OD.QUANTITY) AS QUANTITY_SOLD,
  6         SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_REVENUE
  7  FROM CUSTOMER C
  8  JOIN ORDERS O
  9  ON C.CUSTOMER_ID = O.CUSTOMER_ID
 10  JOIN ORDER_DETAILS OD
 11  ON O.ORDER_ID = OD.ORDER_ID
 12  JOIN PRODUCT P
 13  ON OD.PRODUCT_ID = P.PRODUCT_ID
 14  GROUP BY C.CUSTOMER_ID,
 15           C.CUSTOMER_NAME,
 16           P.PRODUCT_ID,
 17           P.PRODUCT_NAME
 18  ORDER BY TOTAL_REVENUE DESC;

CUSTOMER_I CUSTOMER_NAME                                      PRODUC
---------- -------------------------------------------------- ------
PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD TOTAL_REVENUE
------------- -------------
C005       Karthik                                            P005
Vivo Case
            3          2948

C002       Ananya                                             P002
Samsung Case
            4          2496

CUSTOMER_I CUSTOMER_NAME                                      PRODUC
---------- -------------------------------------------------- ------
PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD TOTAL_REVENUE
------------- -------------

C004       Meena                                              P004
OnePlus Case
            3          1947

C001       Rahul                                              P001
iPhone Case

CUSTOMER_I CUSTOMER_NAME                                      PRODUC
---------- -------------------------------------------------- ------
PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD TOTAL_REVENUE
------------- -------------
            2           948

C003       Vikram                                             P003
Redmi Case
            3           897


SQL>
```

3.least selling product

```
SQL> SELECT P.PRODUCT_ID,
  2         P.PRODUCT_NAME,
  3         SUM(OD.QUANTITY) AS TOTAL_QUANTITY_SOLD
  4  FROM PRODUCT P
  5  JOIN ORDER_DETAILS OD
  6  ON P.PRODUCT_ID = OD.PRODUCT_ID
  7  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  8  HAVING SUM(OD.QUANTITY) = (
  9      SELECT MAX(TOTAL_QUANTITY)
 10      FROM (
 11          SELECT SUM(QUANTITY) AS TOTAL_QUANTITY
 12          FROM ORDER_DETAILS
 13          GROUP BY PRODUCT_ID
 14      )
 15  );

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------
P002
Samsung Case
                  4
```

```
SQL> SELECT P.PRODUCT_ID,
  2         P.PRODUCT_NAME,
  3         SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_REVENUE
  4  FROM PRODUCT P
  5  JOIN ORDER_DETAILS OD
  6  ON P.PRODUCT_ID = OD.PRODUCT_ID
  7  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  8  HAVING SUM(OD.QUANTITY * OD.PRICE) = (
  9      SELECT MAX(PRODUCT_REVENUE)
 10      FROM (
 11          SELECT SUM(QUANTITY * PRICE) AS PRODUCT_REVENUE
 12          FROM ORDER_DETAILS
 13          GROUP BY PRODUCT_ID
 14      )
 15  );

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_REVENUE
-------------
P005
Vivo Case
         2948
```

4.products requiring promotion

```
SQL> SELECT P.PRODUCT_ID,
  2         P.PRODUCT_NAME,
  3         P.PRICE,
  4         P.STOCK_QUANTITY
  5  FROM PRODUCT P
  6  WHERE P.STOCK_QUANTITY > 0
  7  AND NOT EXISTS (
  8      SELECT 1
  9      FROM ORDER_DETAILS OD
 10      WHERE OD.PRODUCT_ID = P.PRODUCT_ID
 11  )
 12  ORDER BY P.PRODUCT_NAME;

no rows selected
```

6.perform category wise sales analysis

1.total sales generated by each category

```
SQL> SELECT C.CATEGORY_ID,
  2         C.CATEGORY_NAME,
  3         SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_SALES
  4  FROM CATEGORY C
  5  JOIN PRODUCT P
  6  ON C.CATEGORY_ID = P.CATEGORY_ID
  7  JOIN ORDER_DETAILS OD
  8  ON P.PRODUCT_ID = OD.PRODUCT_ID
  9  GROUP BY C.CATEGORY_ID, C.CATEGORY_NAME
 10  ORDER BY TOTAL_SALES DESC;

CATEGORY
--------
CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_SALES
-----------
CAT005
Rugged
       2948

CAT002
Leather
       2496

CATEGORY
--------
CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_SALES
-----------

CAT004
Printed
       1947

CAT001
Silicone

CATEGORY
--------
CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_SALES
-----------
        948

CAT003
Transparent
        897
```

2.number of prouducts sold in each category

```
SQL> SELECT C.CATEGORY_ID,
  2         C.CATEGORY_NAME,
  3         NVL(SUM(OD.QUANTITY), 0) AS TOTAL_PRODUCTS_SOLD
  4  FROM CATEGORY C
  5  LEFT JOIN PRODUCT P
  6  ON C.CATEGORY_ID = P.CATEGORY_ID
  7  LEFT JOIN ORDER_DETAILS OD
  8  ON P.PRODUCT_ID = OD.PRODUCT_ID
  9  GROUP BY C.CATEGORY_ID, C.CATEGORY_NAME
 10  ORDER BY TOTAL_PRODUCTS_SOLD DESC;

CATEGORY
--------
CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_PRODUCTS_SOLD
-------------------
CAT002
Leather
                  4

CAT005
Rugged
                  3

CATEGORY
--------
CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_PRODUCTS_SOLD
-------------------

CAT004
Printed
                  3

CAT003
Transparent

CATEGORY
--------
CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_PRODUCTS_SOLD
-------------------
                  3

CAT001
Silicone
                  2
```

3.highest revenue gerating category

```
SQL> SELECT C.CATEGORY_ID,
  2         C.CATEGORY_NAME,
  3         SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_REVENUE
  4  FROM CATEGORY C
  5  JOIN PRODUCT P
  6  ON C.CATEGORY_ID = P.CATEGORY_ID
  7  JOIN ORDER_DETAILS OD
  8  ON P.PRODUCT_ID = OD.PRODUCT_ID
  9  GROUP BY C.CATEGORY_ID, C.CATEGORY_NAME
 10  HAVING SUM(OD.QUANTITY * OD.PRICE) = (
 11      SELECT MAX(CATEGORY_REVENUE)
 12      FROM (
 13          SELECT SUM(OD2.QUANTITY * OD2.PRICE) AS CATEGORY_REVENUE
 14          FROM PRODUCT P2
 15          JOIN ORDER_DETAILS OD2
 16          ON P2.PRODUCT_ID = OD2.PRODUCT_ID
 17          GROUP BY P2.CATEGORY_ID
 18      )
 19  );

CATEGORY
--------
CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_REVENUE
-------------
CAT005
Rugged
         2948
```

4.average sales per category

```
SQL> SELECT C.CATEGORY_ID,
  2         C.CATEGORY_NAME,
  3         NVL(AVG(OD.QUANTITY * OD.PRICE), 0) AS AVERAGE_SALES
  4  FROM CATEGORY C
  5  LEFT JOIN PRODUCT P
  6  ON C.CATEGORY_ID = P.CATEGORY_ID
  7  LEFT JOIN ORDER_DETAILS OD
  8  ON P.PRODUCT_ID = OD.PRODUCT_ID
  9  GROUP BY C.CATEGORY_ID, C.CATEGORY_NAME
 10  ORDER BY AVERAGE_SALES DESC;

CATEGORY
--------
CATEGORY_NAME
--------------------------------------------------------------------------------
AVERAGE_SALES
-------------
CAT005
Rugged
         1474

CAT002
Leather
         1248

CATEGORY
--------
CATEGORY_NAME
--------------------------------------------------------------------------------
AVERAGE_SALES
-------------

CAT004
Printed
        973.5

CAT001
Silicone

CATEGORY
--------
CATEGORY_NAME
--------------------------------------------------------------------------------
AVERAGE_SALES
-------------
          474

CAT003
Transparent
        448.5
```

7 GERATE BUSINESS ANALYTICS REPORTS

Report 1; sales performance report

```
SQL> SELECT SUM(TOTAL_AMOUNT) AS TOTAL_SALES,
  2         COUNT(ORDER_ID) AS NUMBER_OF_ORDERS,
  3         AVG(TOTAL_AMOUNT) AS AVERAGE_ORDER_VALUE,
  4         MAX(TOTAL_AMOUNT) AS HIGHEST_ORDER_VALUE
  5  FROM ORDERS;

TOTAL_SALES NUMBER_OF_ORDERS AVERAGE_ORDER_VALUE HIGHEST_ORDER_VALUE
----------- ---------------- ------------------- -------------------
       7589               10               758.9                1299
```

Report 2; customer analytics report

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.CUSTOMER_NAME,
  3         COUNT(O.ORDER_ID) AS NUMBER_OF_ORDERS,
  4         NVL(SUM(O.TOTAL_AMOUNT), 0) AS TOTAL_SPENDING,
  5         CASE
  6             WHEN COUNT(O.ORDER_ID) >= 3 THEN 'Frequent Customer'
  7             WHEN COUNT(O.ORDER_ID) >= 1 THEN 'Regular Customer'
  8             ELSE 'No Purchases'
  9         END AS PURCHASE_FREQUENCY
 10  FROM CUSTOMER C
 11  LEFT JOIN ORDERS O
 12  ON C.CUSTOMER_ID = O.CUSTOMER_ID
 13  GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
 14  ORDER BY TOTAL_SPENDING DESC;

CUSTOMER_I CUSTOMER_NAME                                      NUMBER_OF_ORDERS
---------- ---------------------------------------------- ----------------
TOTAL_SPENDING PURCHASE_FREQUENC
-------------- -----------------
C004       Meena                                                         2
          1947 Regular Customer

C005       Karthik                                                       2
          1798 Regular Customer

C003       Vikram                                                        2
          1598 Regular Customer


CUSTOMER_I CUSTOMER_NAME                                      NUMBER_OF_ORDERS
---------- ---------------------------------------------- ----------------
TOTAL_SPENDING PURCHASE_FREQUENC
-------------- -----------------
C002       Ananya                                                        2
          1298 Regular Customer

C001       Rahul                                                         2
           948 Regular Customer
```

Report 3; products performance report

```
SQL> SELECT P.PRODUCT_ID,
  2         P.PRODUCT_NAME,
  3         NVL(SUM(OD.QUANTITY), 0) AS QUANTITY_SOLD,
  4         NVL(SUM(OD.QUANTITY * OD.PRICE), 0) AS REVENUE_GENERATED
  5  FROM PRODUCT P
  6  LEFT JOIN ORDER_DETAILS OD
  7  ON P.PRODUCT_ID = OD.PRODUCT_ID
  8  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  9  ORDER BY REVENUE_GENERATED DESC;

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD REVENUE_GENERATED
------------- -----------------
P005
Vivo Case
            3              2948

P002
Samsung Case
            4              2496

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD REVENUE_GENERATED
------------- -----------------

P004
OnePlus Case
            3              1947

P001
iPhone Case

PRODUC
------
PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD REVENUE_GENERATED
------------- -----------------
            2               948

P003
Redmi Case
            3               897
```

Report 4; category analysis report

```
SQL> SELECT C.CATEGORY_ID,
  2         C.CATEGORY_NAME,
  3         NVL(SUM(OD.QUANTITY), 0) AS TOTAL_PRODUCTS_SOLD,
  4         NVL(SUM(OD.QUANTITY * OD.PRICE), 0) AS TOTAL_REVENUE
  5  FROM CATEGORY C
  6  LEFT JOIN PRODUCT P
  7  ON C.CATEGORY_ID = P.CATEGORY_ID
  8  LEFT JOIN ORDER_DETAILS OD
  9  ON P.PRODUCT_ID = OD.PRODUCT_ID
 10  GROUP BY C.CATEGORY_ID, C.CATEGORY_NAME
 11  ORDER BY TOTAL_REVENUE DESC;

CATEGORY
--------
CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_PRODUCTS_SOLD TOTAL_REVENUE
------------------- -------------
CAT005
Rugged
                  3          2948

CAT002
Leather
                  4          2496

CATEGORY
--------
CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_PRODUCTS_SOLD TOTAL_REVENUE
------------------- -------------

CAT004
Printed
                  3          1947

CAT001
Silicone

CATEGORY
--------
CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_PRODUCTS_SOLD TOTAL_REVENUE
------------------- -------------
                  2           948

CAT003
Transparent
                  3           897
```

```
SQL> SELECT C.CATEGORY_NAME,
  2         SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_SALES
  3  FROM CATEGORY C
  4  JOIN PRODUCT P
  5  ON C.CATEGORY_ID = P.CATEGORY_ID
  6  JOIN ORDER_DETAILS OD
  7  ON P.PRODUCT_ID = OD.PRODUCT_ID
  8  GROUP BY C.CATEGORY_NAME
  9  ORDER BY TOTAL_SALES DESC;

CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_SALES
-----------
Rugged
       2948

Leather
       2496

Printed
       1947


CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_SALES
-----------
Silicone
        948

Transparent
        897


SQL>
SQL>
```